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EXTENSION SERVICE

COLLEGE OF AGRICULTURE UNIVERSITY OF ARKANSAS
U. S. DEPARTMENT OF AGRICULTURE
COOPERATING

Home-Made Soap

MRS. IDA A. FENTON

EXTENSION CIRCULAR NO. 305

DAN T. GRAY, Director
C. C. RANDALL, Acting Assistant Director

REPRINT, JUNE, 1936

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EXTENSION CIRCULARS

The Extension Service, College of Agriculture University of Arkansas, issues a large number of circulars on a wide variety of farm and home practices as a service to the farm people of Arkansas. Among the available Extension Circulars on home management are:

- No. 174 Arrangement of Kitchen Equipment
- No. 211 Low-Cost Farm Water Systems
- No. 245 Refinishing Furniture
- No. 252 Floors, Their Finish and Care
- No. 284 Standard Small Equipment for the Kitchen
- No. 290 Management of Stoves
- No. 339 Refinishing Old Oil Stoves
- No. 342 Housekeeping Made Easy
- No. 349 Improved Methods of Home Laundry
- No. 362 Remedies and Preventives for Household Pests

Extension Circulars on other subjects, as well as those listed here, may be secured from your County or Home Demonstration Agent.

HOME-MADE SOAP

MRS. IDA A. FENTON

Extension Economist—Home Management

The average Arkansas family will require approximately \$10 worth of soap during a year. This is estimated as follows:

52 weeks	{	2 bars per week for laundering	
		1 bar per week for dishes and kitchen cleaning	
		1/3 bar per week for special cleaning	
		1¼ bars per week for toilet purposes	
One hundred seventy-three bars for laundering and cleaning at 4 cents			\$ 6.92
Sixty-five bars for toilet purposes at 5 cents			3.25
Total			\$10.17

This article from the household budget may be reduced 85 per cent by utilizing the waste fat from the home.

DIRECTIONS FOR MAKING SOAP

A good soap is a combination of fat, alkali (usually lye), and water. When properly combined the soap contains practically no free alkali, is not greasy, and is of such a consistency that the bar will hold its shape and not crumble. Such a soap can be made in the home by following simple directions. It is best made by the cold process. Experience in making soap is not necessary for success with this method.

TO SWEETEN RANCID FAT

Boil the fat in 1 part vinegar and 5 parts water, and allow to cool; then skim off the fat.

TO RID FAT OF IMPURITIES

Boil the fat in water and let it stand until it is cold; then skim off the fat. Meat fryings must be so treated to rid them of salt.

PREPARATION OF CRACKLINGS, MEAT RINDS, ETC., FOR SOAP MAKING

To each gallon, or 4 pounds of rinds and other waste fat, add 1 tablespoon of lye, and water to twice the depth of the cracklings, and cover and boil for 1 hour. Remove from the fire and when boiling stops, pour sufficient cold water over it so that fat may float free; when cold skim off the fat. If the fat is not entirely free from foreign substances, treat as for ridding fat of impurities.

SELECTION OF FAT FOR SOAP

Half lard and half tallow gives a good laundry soap. Half lard or vegetable oil and half tallow is better for toilet soaps.

WATER CONTENT

Use rain water if possible. If hard water is used let it stand 2 or 3 days with a small amount of lye (from a teaspoon to 2 tablespoons to a gallon, depending upon the hardness of the water), so that all hardness may settle. If a neutral soap is desired test the water with red litmus paper until a red paper remains. If it turns blue add more water.

MEASUREMENTS

Accurate measurements are essential to success.

TEMPERATURE

Correct temperature assures success. The following chart will be helpful. A dairy or floating thermometer may be used. Temperatures of fat and lye vary with room temperatures. In a 70-degree room temperature, use the following:

Lard (soft fat) 100° F.; lye 70° F.

Tallow (hard fat) 130° F.; lye 90° F.

Tallow and lard mixture 115° F.; lye 80° F.

Soft rancid fat 98° F.; lye 77° F.

In a combination of hard and soft fats, use a temperature between the specified temperature for each. In hot weather the temperature of fat may be about 10 degrees less and lye from 2 to 4 degrees less.

MANIPULATION

Manipulation is important; stir gently in one direction only. Gentle pouring is also important. Soap should be cured at least 2 weeks. This allows free lye to saponify. Soap made of lard or oil requires longer than others. Toilet soap requires a month to cure. Do not allow soap to freeze.

Basic formula (yields 9 pounds of soap):

6 pounds fat

2½ pints water

1 can high grade lye

Dissolve the lye in the water, using a stone, glass, or enameled container. Let it cool to the proper temperature. Melt the fat in an enameled pan and stir with a wooden paddle. Bring to

the correct temperature. Stir to be certain the fat in center of pan is the same temperature as fat at the edges of the pan.

Pour the lye solution into the fat in a thin, steady stream with slow, even stirring. A honey-like texture is formed and in about 10 to 20 minutes it becomes thick with all of the lye incorporated in the fat. Pour into a wooden box that has been soaked in water and lined with clean cotton cloth dipped in water and wrung out nearly dry. Cover with a board or cardboard and then with a rug or blanket to retain the heat. Let it remain undisturbed for 24 hours. To remove from the mould lift by the ends of overhanging cloth. Mark off into bars and cut by wrapping a stout string or fine wire around the soap cake once; cross the ends and pull. Place the soap so air can reach it, but avoid drafts and cold.

TOILET SOAP

Use the same basic formula and proceed as directed. Sweet lard and tallow should be used for toilet soap, never fat that has been used for cooking or has become rancid.

VARIATIONS IN SOAP MAKING

FLOATING SOAP

When the soap mixture has reached the honey stage fold air into it as eggwhite is folded into a cake mixture.

PERFUMED SOAP

The following oils are recommended:

Oil of citronella—2 teaspoons to basic formula (cheap and pleasant)

Oil of wintergreen—2 teaspoons to basic formula

Oil of lavender—2 teaspoons to basic formula

Oil of bergamot—2 teaspoons to basic formula

Oil of sassafras—4 teaspoons to basic formula

Oil of rosemary—2 teaspoons to basic formula

Oil of rose geranium— $\frac{1}{2}$ teaspoon to basic formula

These can be secured at a drug store.

COLOR

The following colors are non-poisonous and non-fading:

Yellow—Fluorescein (5 grains in one ounce of water)

Green—Naphthol green (b) (4 grains in 1 ounce of water)

Red—Rhodamine B (1 grain in 1 ounce of water)

These can be secured from a drug store. They will cost from a half cent to one cent per pound of soap.

BORAX SOAP

Borax will quicken the sudsing action of soap. Two table-spoons may be added to the lye solution while cooling.

FOR EXTREMELY SENSITIVE OR DRY SKIN TOILET SOAP

The addition of six ounces of glycerine to the fat may be desirable for such persons.

RECLAIMING SOAP

If at the end of 24 hours there is a film of grease on top, leave for 48 hours until the grease disappears and then cut. If there is a liquid on the bottom, cut the soap into small pieces and let stand until all liquid is absorbed. If, after standing, liquid is not absorbed, shave soap into kettle, add liquor from box and 7 pints of water. Stir slowly and evenly and bring to a boil. In a few minutes they will blend, then pour gently into a box. The bulk will be greater but the cleansing properties will be the same.

SOAP FLAKES

When soap has cured from 2 to 4 weeks shave by rubbing bars over a vegetable slicer or a kraut cutter.

NOTES

Don't use aluminum utensils.

Separation is caused by too much heat, too much cold, rough handling, or salt in the fat.

Excess lye makes a coarse, flinty soap; it crumbles when shaved.

Mineral oils will not make soap.

